THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE UNIT TEST SET 3

Chapters 6–10: **We Distribute, Yet Things Multiply, Proportional Reasoning-1, Fractions in Disguise, The Baudhayana–Pythagoras Theorem, Proportional Reasoning-2** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Which of the following is true for all values of a and b? 1

(a) $(a-b)^2 = a^2 - 2ab + b^2$ **(b)** $(a+b)^2 = a^2 + b^2$ **(c)** $(a+b)(a-b) = a^2 + b^2$ **(d)** $(a-b)^2 = a^2 - b^2$

2. If the number of workers increases and the job remains the same, the time taken: 1

(a) becomes zero **(b)** decreases **(c)** increases **(d)** stays the same

3. A value changes from 250 to 200. The percentage decrease is: 1

(a) 22% **(b)** 20% **(c)** 18% **(d)** 40%

4. The Baudhayana–Pythagoras theorem states that in a right triangle: 1

(a) the hypotenuse is the shortest side
(b) the sum of the two legs equals the hypotenuse
(c) the square of the hypotenuse equals the sum of the squares of the other two sides
(d) the square of a leg equals the sum of the other two sides

5. Which of the following is the greatest ratio? 1

(a) 5:8 **(b)** 4:7 **(c)** 7:12 **(d)** 3:5

6. $(x+4)^2$ equals: 1

(a) $x^2 + 8x$ **(b)** $x^2 + 4x + 16$ **(c)** $x^2 + 16$ **(d)** $x^2 + 8x + 16$

7. The ratio of 1 m 50 cm to 30 cm is: 1

(a) 5 : 2 **(b)** 6 : 1 **(c)** 5 : 1 **(d)** 1 : 5

8. 35 is what percent of 140? 1

(a) 25% **(b)** 28% **(c)** 50% **(d)** 22%

9. The legs of a right triangle are 2 cm and 6 cm. Its hypotenuse (in cm) is: 1

(a) $2\sqrt{10}$ **(b)** 8 **(c)** $2\sqrt{2}$ **(d)** $\sqrt{41}$

10. If A:B = 3:4 and B:C = 5:6, then A:B:C is: 1

(a) 3:4:6 **(b)** 3:9:6 **(c)** 15:24:10 **(d)** 15:20:24

11. $(x-5)^2$ equals: 1

(a) $x^2 + 10x + 25$ **(b)** $x^2 - 10x + 25$ **(c)** $x^2 - 5x + 25$ **(d)** $x^2 - 25$

12. If A : B = 3 : 4 and B : C = 8 : 7, then A : B : C is: 1

(a) 6 : 8 : 7 **(b)** 3 : 4 : 7 **(c)** 3 : 8 : 7 **(d)** 3 : 4 : 56

13. A cycle bought for ₹ 2400 is sold for ₹ 2160. The loss percent is: 1

(a) 20% **(b)** 10% **(c)** 11% **(d)** 9%

14. The legs of a right triangle are 11 cm and 13 cm. The area of the square drawn on its hypotenuse (in cm²) is: 1

(a) 290 **(b)** 576 **(c)** 24 **(d)** 143

15. 20 men can do a piece of work in 9 days. 15 men will do it in (days): 1

(a) 12 **(b)** 10 **(c)** 180 **(d)** 15

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Find 48×52 using $(a+b)(a-b)$. 2

17. A map scale is 1 cm = 20 km. Find the actual distance for 3.5 cm and the map distance for 100 km. 2

18. Find the rate of simple interest if ₹ 4000 becomes ₹ 4720 in 3 years. 2

19. Find the length of the diagonal of a square of side 4 cm. 2

20. y is directly proportional to x. If $y = 35$ when $x = 7$, find the constant of proportionality and y when $x = 12$. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A school bus travelling at 30 km/h takes 24 minutes to reach the school. At what speed must it travel to reach in 18 minutes? 3

22. Expand $(3x+4)^2$. 3

23. If A : B = 3 : 4 and B : C = 8 : 7, find A : B : C. 3

24. The marked price of a laptop is ₹ 36000. GST at 12% is added. Find the GST amount and the final bill. 3

OR

The hypotenuse of an isosceles right triangle is $6\sqrt{2}$ cm. Find the length of each equal side and the area of the triangle. 3

25. On a map of scale 1 cm = 3 km, a rectangular park measures 6 cm by 5 cm. Find the actual length, breadth and perimeter of the park. 3

OR

Find 68×72 using the identity $(a+b)(a-b) = a^2 - b^2$. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Two poles of heights 20 m and 12 m stand on level ground 15 m apart. Find the distance between their tops. 5

OR

The marks of a student are: Maths 18, Science 15, English 9, Hindi 12, SST 6. Total 60. Find the central angle of every sector of a pie chart and the percentage of the total taken by Maths. 5

27. A rectangle has length $(x + 2)$ cm and breadth $(x + 6)$ cm. Write an expression for its area and find the area when $x = 5$. 5

OR

Divide ₹ 1800 among A, B and C in the ratio 2 : 3 : 4. If C gives ₹ 200 to A, find the new ratio of their amounts. 5

END OF QUESTION PAPER **Total: 50 Marks**

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